

AML / COMPLIANCE TAXONOMY AUDIT

PART 3 – FINAL TAXONOMY

FAMILY AML-001: AI-ENHANCED ADAPTIVE TRANSACTION MONITORING EVASION

Identity

Field	Value
Attack ID	AML-001
Attack Family	AI-Enhanced Adaptive Transaction Monitoring Evasion
Variants	Pattern-free movement; Behavioral camouflage; Risk-score manipulation; Threshold-conscious structuring; Adaptive transaction pattern generation; Temporal pattern optimization; Velocity rule evasion
Primary Lifecycle Stage	AML / Compliance
Cross-stage Stages	Authorization (score manipulation); Payment Initiation (pattern generation); Cash-out/Mule (movement); Settlement (timing)
Attacker	Sophisticated money launderers; Organized crime groups; Fraud rings with AI capability
Target	Transaction monitoring systems; ML-based anomaly detection; Behavioral analytics; Velocity rules; Risk scoring models
Objective	Conduct fraudulent activity without triggering AML transaction monitoring alerts, maintaining a low risk profile, avoiding enhanced scrutiny, and evading detection over sustained periods
AML Control Targeted	Transaction monitoring rules; ML-based anomaly detection; Behavioral analytics; Velocity rules; Alert thresholds; Risk scoring

Fraud Understanding

What is the attack?

Attackers use GenAI to generate transaction patterns that are statistically indistinguishable from legitimate activity, systematically evading ML-based transaction monitoring and behavioral analytics. Unlike traditional structuring (which relies on static threshold avoidance), GenAI enables adaptive, ML-evasive pattern generation that evolves in real-time based on detection feedback. The attack mimics "normal" transaction behavior, randomizes transfer timing to avoid detection patterns, splits transactions to stay under alert thresholds, and adapts to dodge rule-based security systems.

Traditional mechanism:

Before GenAI, attackers evaded AML monitoring through manual transaction structuring (breaking large transactions into smaller amounts), volume smoothing, low-and-slow approaches, beneficiary diversification, and scheduling fraud during off-hours. These methods were detectable by ML-based monitoring because they relied on static patterns or unnatural consistency.

GenAI transformation:

Dimension	Pre-GenAI	GenAI-Transformed
Evasion approach	Static, rule-based avoidance	Adaptive, ML-based evasion
Pattern generation	Manual structuring	AI-optimized transaction patterns
Adaptation	None	Real-time feedback adaptation
Scale	Limited	Thousands of optimized transactions
Detection evasion	Avoids known thresholds	Evades ML-based monitoring
Behavioral mimicry	Limited	Indistinguishable from legitimate

HARD test result: PASS — GenAI creates a qualitatively new capability. The transformation from manual, detectable rule evasion to adaptive, ML-evasive pattern generation is a qualitative leap.

Prerequisites:

- Existing accounts (legitimate or fraudulently obtained)

- Understanding of AML monitoring parameters (can be probed)
- Ability to control transaction patterns
- GenAI capability for pattern generation and optimization
- (For adaptive variant) Access to detection feedback

Conceptual attack flow:

1. Attacker studies AML monitoring rules, thresholds, and ML model behavior
2. Attacker establishes baseline understanding of detection parameters through probing
3. Attacker uses GenAI to generate optimized transaction patterns with natural variation
4. Attacker executes transactions at levels/scoring that appear normal
5. Risk score remains low; no alerts generated
6. Fraud continues undetected over time
7. If monitoring changes, GenAI adapts patterns accordingly
8. Detection is delayed until significant losses have accumulated

Consequence:

Fraud continues undetected over time; cumulative losses accumulate; delayed detection allows funds to be laundered before intervention; regulatory penalties for institution.

Detection Intelligence

Signal Category	Observable Signal	Why It Matters	Detection Approach	FP Risk
Transaction	Amounts just below thresholds with unnatural consistency	Indicates threshold-conscious optimization	Rules (tolerance bands); ML anomaly detection	Moderate
Transaction	Volume patterns with unnatural smoothness	Optimization creates unnatural consistency	Anomaly detection (Isolation Forest, Autoencoder)	Moderate
Transaction	Beneficiary patterns with unnatural rotation	Optimization creates unnatural patterns	Pattern analysis; Graph analytics	Moderate
Account	Behavioral patterns that are "too perfect"	Optimization creates unnatural consistency	Behavioral analytics	High
Account	Lack of natural behavioral drift	Absence of natural variation indicates optimization	Sequence modeling (LSTM/Transformer)	High
Entity	Business profiles with perfect consistency	Too-perfect alignment indicates optimization	Rules + ML	High
Network	Coordinated patterns across accounts	Indicates shared optimization strategy	Graph Neural Network; Clustering	Low
Risk	Risk score unnatural stability	Successful evasion keeps scores consistently low	ML anomaly detection	Moderate
Temporal	Growth patterns with unnatural smoothness	Unnatural smoothness in growth	Temporal analysis	Moderate
Behavioral	Patterns that evolve in response to detection changes	Indicates feedback-driven optimization	Temporal pattern analysis	Low

Simulation Intelligence

Component	Description	Classification
Legitimate entities	Account personas with business profiles, transaction histories	Algorithmic
Suspicious entities	Accounts with optimized transaction patterns	Algorithmic + Agentic
Transactions	Optimized amounts, timing, velocities with natural variation	Algorithmic
Account histories	Historical legitimate patterns (baseline)	Algorithmic
Entity relationships	Account-merchant relationships; beneficiary networks	Algorithmic
Transaction graphs	Flow-of-funds patterns	Algorithmic
Behavioral timelines	Pattern evolution over 30-365 days	Agentic
Attack labels	Monitoring evasion; Threshold optimization; Pattern camouflage	Algorithmic
Mutation parameters	Pattern variation; Adaptation rate; Noise injection; Strategy shifts	Algorithmic
Legitimate counterparts	Well-managed businesses; Subscription billing; Seasonal businesses	Algorithmic

Component	Description	Classification
Edge cases	Partial evasion; Mixed legitimate/fraudulent patterns; Portfolio-level evasion	Hybrid

Simulation Type: Agentic + Algorithmic

Evidence

Claim	Source	Publisher	Date	Evidence Type	Confidence	Maturity
AI-generated transaction profiles can evade detection	Qredible	Qredible	2025	Industry analysis	VERIFIED	EMERGING
Legacy controls built for human-speed scams now outmatched	Visa Threats Report	Visa	2025	Payment network	VERIFIED	STRONG RESEARCH
AI-generated content can be indistinguishable from legitimate materials	Visa	Visa	2025	Payment network	VERIFIED	STRONG RESEARCH
Traditional scoring models are showing a weak spot	TransUnion	TransUnion	2025	Industry report	VERIFIED	STRONG RESEARCH
Criminals use AI to optimize transaction patterns that evade detection	Payment Initiation Taxonomy	Research	2025	Industry analysis	REASONABLE INFERENCE	EMERGING
High-frequency, low-value "micro frauds" can quietly erode up to ~2% of GMV	Payment Initiation Taxonomy	Research	2025	Industry analysis	REASONABLE INFERENCE	EMERGING
AI mimics "normal" transaction behavior and adapts to evade detection	Overwatch Data	Overwatch	2025	Industry analysis	VERIFIED	EMERGING
Criminals increasingly exploit GenAI to scale illicit finance	FATF October 2025 Plenary	FATF	2025	Government report	VERIFIED	STRONG RESEARCH

FAMILY AML-002: AI-POWERED AML ALERT FLOODING / DOS

Identity

Field	Value
Attack ID	AML-002
Attack Family	AI-Powered AML Alert Flooding / DoS
Variants	False-positive flooding; Alert system overload; Alert fatigue exploitation; Investigator overwhelm; Synthetic alert generation; Mass false-positive generation
Primary Lifecycle Stage	AML / Compliance
Cross-stage Stages	Account Creation (synthetic accounts); Payment Initiation (multiple instructions)
Attacker	Sophisticated money launderers; Organized crime groups; State actors; Fraud rings
Target	AML alert generation systems; Case management workflows; Compliance investigators; Alert investigation queues
Objective	Overwhelm AML systems with massive volumes of AI-generated suspicious activity alerts, causing alert fatigue, delayed investigation of genuine threats, and system degradation
AML Control Targeted	Alert generation; Alert triage; Case management; Investigation workflows; Investigator attention

Fraud Understanding

What is the attack?

Attackers use GenAI to generate massive volumes of synthetic suspicious activity patterns that trigger AML alerts—without actually executing fraud. The goal is to overwhelm the AML system's alert generation, triage, and investigation capacity, causing alert fatigue and delayed response to genuine threats. The FATF has noted that criminals may create thousands of synthetic identities to overwhelm detection systems.

Traditional mechanism:

Before GenAI, attackers could not generate sufficient volume of varied suspicious patterns to overwhelm AML systems. The attack required manual generation of suspicious activity, which was limited in scale and detectable as coordinated.

GenAI transformation:

Dimension	Pre-GenAI	GenAI-Transformed
Alert generation volume	Limited by manual effort	Industrial-scale, automated
Pattern variety	Limited, detectable	High variety, indistinguishable
Coordination	Manual	Autonomous, adaptive
Scale	Dozens per campaign	Thousands per campaign
Detection of flooding	Detectable as coordinated	Appears as legitimate variation

HARD test result: PASS – GenAI creates a qualitatively new capability. The transformation from limited, detectable flooding to industrial-scale, adaptive alert generation is a qualitative leap.

Prerequisites:

- Access to synthetic accounts (or ability to generate patterns)
- Understanding of AML alert generation rules
- Ability to generate transaction patterns that trigger alerts
- GenAI capability for pattern variation at scale
- (For scale) AI agent infrastructure

Conceptual attack flow:

1. Attacker studies AML alert generation rules and thresholds
2. Attacker uses GenAI to generate transaction patterns that will trigger alerts
3. Attacker executes large-scale pattern generation across multiple accounts
4. AML system generates massive volume of alerts
5. Alert queue becomes overwhelmed
6. Investigators experience alert fatigue; genuine alerts may be missed
7. Genuine fraud executed through other accounts while AML resources are consumed

Consequence:

Alert fatigue among investigators; delayed response to genuine threats; missed suspicious activity; system performance degradation; increased operational costs; regulatory penalties for failure to investigate.

Detection Intelligence

Signal Category	Observable Signal	Why It Matters	Detection Approach	FP Risk
Alert	Sudden spike in alert volume	Indicates flooding attack	Volume anomaly detection	Moderate
Alert	High ratio of false positives	Indicates synthetic generation	Alert outcome analysis	Low
Alert	Multiple accounts generating similar alert patterns	Indicates coordinated flooding	Clustering; Graph analytics	Low
Transaction	Patterns that exactly match alert rules	Indicates intentional triggering	Rules + ML	Moderate
Transaction	Unnatural pattern variety	Indicates AI generation	ML anomaly detection	Moderate
Network	Account relationships in flooding campaign	Indicates coordinated attack	Graph Neural Network	Low
Temporal	Coordinated alert timing across accounts	Indicates orchestrated attack	Temporal correlation	Low
Investigation	Decreasing investigation quality metrics	Alert fatigue indicator	Operational analytics	Low
Investigation	Increasing time-to-investigate	System overload indicator	Operational analytics	Low

Simulation Intelligence

Component	Description	Classification
Legitimate entities	Account personas with normal transaction patterns	Algorithmic
Suspicious entities	Accounts generating alert-triggering patterns	Algorithmic + Agentic
Transactions	Patterns designed to trigger specific alert types	Algorithmic
Account histories	Accounts with varying historical patterns	Algorithmic
Alert sequences	Sequences of alerts with varying characteristics	Agentic
Behavioral timelines	Alert generation patterns over time	Agentic
Attack labels	Alert flooding; False-positive generation	Algorithmic

Component	Description	Classification
Mutation parameters	Pattern variation; Timing variation; Alert type variation	Algorithmic
Legitimate counterparts	Genuine suspicious activity increases	Algorithmic
Edge cases	Mixed legitimate/fraudulent alert patterns	Hybrid

Simulation Type: Agentic + Algorithmic

Evidence

Claim	Source	Publisher	Date	Evidence Type	Confidence	Maturity
Criminals may create thousands of synthetic identities to overwhelm detection systems	FATF Horizon Scan	FATF	Dec 2025	Government report	VERIFIED	STRONG RESEARCH
AI can generate synthetic data to "poison" ML detection models	FATF Horizon Scan	FATF	Dec 2025	Government report	VERIFIED	STRONG RESEARCH
Alert fatigue is a recognized problem in AML operations	ACAMS	ACAMS	2025	Industry analysis	VERIFIED	STRONG RESEARCH
Adversarial ML attacks can target ML-based fraud detection	Mastercard	Mastercard	2025	Payment network	VERIFIED	STRONG RESEARCH

FAMILY AML-003: AI-GENERATED SYNTHETIC AML NARRATIVE / EVIDENCE

Identity

Field	Value
Attack ID	AML-003
Attack Family	AI-Generated Synthetic AML Narrative / Evidence
Variants	Synthetic transaction narratives; Fabricated business justification; AI-generated customer communications; Fabricated supporting documentation; Synthetic financial histories; AI-generated source of funds explanations
Primary Lifecycle Stage	AML / Compliance
Cross-stage Stages	Account Creation (documentation); Payment Initiation (narrative generation); Investigation (evidence fabrication)
Attacker	Money launderers; Fraudsters; Sanctions evaders; Organized crime groups
Target	AML investigation workflows; Customer due diligence; Source of funds verification; Compliance decisioning; Regulatory reporting
Objective	Generate convincing synthetic narratives, justifications, and evidence to explain suspicious activity, evade investigation, and comply with AML inquiries
AML Control Targeted	Investigation workflows; CDD/EDD; Source of funds verification; Regulatory reporting; Compliance decisioning

Fraud Understanding

What is the attack?

Attackers use GenAI to generate convincing synthetic narratives, justifications, and supporting evidence to explain suspicious activity and evade AML investigation. When transactions are flagged for review, AI-generated explanations can provide internally consistent, plausible justifications that satisfy investigator requirements. Research demonstrates that LLMs deployed in financial contexts are vulnerable to multi-turn adversarial attacks utilizing a "Risk-Concealment" strategy.

Traditional mechanism:

Before GenAI, attackers manually created explanations or used template-based justifications that were often inconsistent or detectable. Limited personalization made them easier for investigators to identify as fraudulent.

GenAI transformation:

Dimension	Pre-GenAI	GenAI-Transformed
Narrative quality	Generic, inconsistent	Internally consistent, context-aware

Dimension	Pre-GenAI	GenAI-Transformed
Personalization	Limited	Hyper-personalized to the account
Scale	Manual	Automated at scale
Detection evasion	Detectable inconsistencies	Passes consistency checks
Supporting documents	Crude forgeries	Professional, photorealistic
Adaptation	Static	Adaptive to investigation questions

HARD test result: PASS – GenAI creates a qualitatively new capability. The transformation from generic, inconsistent explanations to hyper-personalized, internally consistent narratives is a qualitative leap.

Prerequisites:

- Account flagged for investigation
- Understanding of AML investigation process
- Access to GenAI tools (LLMs for narrative, image generation for documents)
- Knowledge of the account's transaction history

Conceptual attack flow:

1. Account activity triggers AML alert
2. Attacker receives notice of investigation (or anticipates it)
3. Attacker uses GenAI to generate plausible explanation
4. Narrative incorporates account history, transaction patterns, business context
5. Attacker submits explanation to compliance team
6. Investigation reviews narrative; appears internally consistent
7. Investigation closed or findings mitigated
8. Fraud activity continues undetected

Consequence:

Fraud activity continues undetected; regulatory filings inaccurate; investigation resources wasted; increased exposure to financial crime.

Detection Intelligence

Signal Category	Observable Signal	Why It Matters	Detection Approach	FP Risk
Documentation	AI-generated text patterns in explanations	Indicates synthetic generation	NLP; LLM-based detection; Perplexity analysis	High
Documentation	Unnatural consistency across documents	Indicates AI generation	Cross-document consistency scoring	High
Documentation	Metadata anomalies	Indicates fabrication	Metadata analysis	Moderate
Documentation	Missing or inconsistent security features	Indicates forgery	Document forensics	Moderate
Behavioral	Unnatural response timing	Indicates automated generation	Temporal analysis	Low
Behavioral	Too-perfect responses	Indicates AI generation	NLP analysis	High
Network	Similar narrative patterns across accounts	Indicates coordinated fabrication	Clustering	Low

Simulation Intelligence

Component	Description	Classification
Legitimate entities	Account personas with legitimate transaction histories	Algorithmic
Suspicious entities	Accounts under investigation with synthetic explanations	Agentic
Transactions	Transaction patterns being explained	Algorithmic
Account histories	Historical transaction data	Algorithmic
Narratives	AI-generated transaction explanations; Business justifications	Agentic
Documents	Invoices; Contracts; Business agreements; Financial statements	Agentic + Algorithmic
Investigation interactions	Attacker-investigator communications	Agentic
Attack labels	Narrative fabrication; Investigation evasion	Algorithmic

Component	Description	Classification
Mutation parameters	Narrative variation; Document variation; Personalization level	Algorithmic
Legitimate counterparts	Legitimate business documentation; Professional communications	Algorithmic
Edge cases	Mixed real/fabricated information; Partial fabrication	Hybrid

Simulation Type: Agentic (primary) + Algorithmic (document generation)

Evidence

Claim	Source	Publisher	Date	Evidence Type	Confidence	Maturity
LLMs vulnerable to multi-turn adversarial "Risk-Concealment" strategy for money laundering	Academic research	Research	2025	Academic research	VERIFIED	STRONG RESEARCH
Malicious actors can obtain sophisticated strategies for money laundering through LLM-based adversarial attacks	Academic research	Research	2025	Academic research	VERIFIED	STRONG RESEARCH
Deepfakes allow fabrication of documentary evidence to support fraudulent firms or sham transactions	FATF	FATF	2025	Government report	VERIFIED	STRONG RESEARCH

FAMILY AML-004: GENAI-ENHANCED SANCTIONS / PEPs SCREENING EVASION

Identity

Field	Value
Attack ID	AML-004
Attack Family	GenAI-Enhanced Sanctions / PEPs Screening Evasion
Variants	Identity obfuscation; Name variation generation; Entity structure concealment; Beneficial ownership concealment; Sanctions evasion through synthetic entities; PEP status concealment
Primary Lifecycle Stage	AML / Compliance
Cross-stage Stages	Identity/KYC (entity creation); Account Creation (onboarding); Payment Initiation (sanctions evasion)
Attacker	Sanctions evaders; Proliferation financiers; Terrorist financiers; Corrupt officials; Organized crime groups
Target	Sanctions screening systems; PEP screening; Entity resolution; Beneficial ownership verification; Watchlist monitoring
Objective	Evade sanctions and PEP screening by obfuscating identity, concealing beneficial ownership, or generating synthetic entities that bypass watchlist matches
AML Control Targeted	Sanctions screening; PEP screening; Entity resolution; Watchlist matching; Beneficial ownership verification

Fraud Understanding

What is the attack?

Attackers use GenAI to generate variations of names, entity structures, and beneficial ownership arrangements that evade sanctions and PEP screening systems. By generating plausible but varied identities and structures, attackers can create entities that appear legitimate while concealing the true sanctioned or politically exposed nature of the ultimate beneficial owner. RUSI's commentary is blunt: "The era of AI as an 'emerging technology' is over. AI-enabled sanctions evasion is already here."

Traditional mechanism:

Before GenAI, sanctions evasion relied on manual identity variation, complex corporate structures, and jurisdictional arbitrage. The scale and sophistication of evasion was limited by human effort and the ability to generate convincing documentation.

GenAI transformation:

Dimension	Pre-GenAI	GenAI-Transformed
Identity variation	Manual, limited	AI-generated variations at scale
Entity structure	Manual creation	AI-optimized structures
Documentation	Forged, detectable	AI-generated, professional

Dimension	Pre-GenAI	GenAI-Transformed
Detection evasion	Limited	Adaptive to screening rules
Scale	Limited	Industrial-scale

HARD test result: PASS — GenAI creates a qualitatively new capability. The transformation from manual, detectable evasion to adaptive, AI-generated evasion is a qualitative leap.

Prerequisites:

- Knowledge of sanctions screening logic
- Understanding of watchlist matching
- Access to GenAI tools for identity/document generation
- (For scale) Access to synthetic identity infrastructure

Conceptual attack flow:

1. Attacker identifies target sanctions watchlist
2. Attacker uses GenAI to generate identity variations that evade matching
3. Attacker creates synthetic entity structure with fabricated documentation
4. Entity submitted for onboarding
5. Sanctions screening passes (no match found)
6. Account created under synthetic entity
7. Funds moved through entity to sanctioned parties

Consequence:

Sanctions violations; regulatory penalties; reputational damage; funds flow to sanctioned parties; systemic AML/CFT risk.

Detection Intelligence

Signal Category	Observable Signal	Why It Matters	Detection Approach	FP Risk
Identity	Unnatural name variations	Indicates evasion attempt	ML anomaly detection; Fuzzy matching	High
Identity	Inconsistent identity details	Indicates synthetic generation	Entity resolution	Moderate
Entity	Unnatural corporate structure	Indicates evasion attempt	Graph analytics	High
Entity	Jurisdictional patterns	Indicates evasion attempt	Geographic analysis	Moderate
Documentation	AI-generated document artifacts	Indicates fabrication	Document forensics	Moderate
Network	Connections to known sanctioned entities	Indicates evasion	Graph Neural Network	Low
Network	Entity relationship patterns	Indicates beneficial ownership concealment	Graph analytics	Low
Temporal	Rapid entity creation patterns	Indicates industrial-scale evasion	Velocity rules	Moderate

Simulation Intelligence

Component	Description	Classification
Legitimate entities	Entity personas with legitimate identity attributes	Algorithmic
Suspicious entities	Entities designed to evade screening	Algorithmic + Agentic
Identity variations	Name variations; Identity details	Algorithmic
Entity structures	Corporate structures; Beneficial ownership arrangements	Agentic
Documents	Incorporation certificates; Identity documents	Algorithmic
Entity relationships	Ownership chains; Control relationships	Algorithmic
Attack labels	Sanctions evasion; PEP evasion; Entity concealment	Algorithmic
Mutation parameters	Name variation; Structure variation; Documentation variation	Algorithmic
Legitimate counterparts	Legitimate name variations; Legitimate corporate structures	Algorithmic
Edge cases	Partial evasion; Mixed legitimate/concealed structures	Hybrid

Evidence

Claim	Source	Publisher	Date	Evidence Type	Confidence	Maturity
AI-enabled sanctions evasion is already here	RUSI	RUSI	2025	Research commentary	VERIFIED	STRONG RESEARCH
AI can generate entire fake clients, documents and corporate networks at scale	DFAT	DFAT	2025	Government guidance	VERIFIED	STRONG RESEARCH
Criminals use AI to generate realistic fraudulent documents, build and manage networks of synthetic entities	RUSI/FATF/DFAT	RUSI/FATF/DFAT	2025	Government/Research	VERIFIED	STRONG RESEARCH
FATF warns criminals exploit GenAI to scale illicit finance	FATF October 2025 Plenary	FATF	2025	Government report	VERIFIED	STRONG RESEARCH
AI automates complex shell-company structures and trade patterns	FATF June 2025 report	FATF	2025	Government report	VERIFIED	STRONG RESEARCH

FAMILY AML-005: AI-POWERED AML INVESTIGATOR SOCIAL ENGINEERING

Identity

Field	Value
Attack ID	AML-005
Attack Family	AI-Powered AML Investigator Social Engineering
Variants	Investigator impersonation; AI-generated investigation responses; Compliance workflow manipulation; Evidence fabrication; Case manipulation
Primary Lifecycle Stage	AML / Compliance
Cross-stage Stages	Employee/Insider (targeting); Investigation (manipulation); Reporting (SAR manipulation)
Attacker	Sophisticated money launderers; Organized crime groups; Fraud rings; External attackers
Target	AML investigators; Compliance officers; Case management systems; Investigation workflows
Objective	Manipulate AML investigators or investigation workflows to close suspicious activity cases, downgrade risk assessments, or manipulate regulatory filings
AML Control Targeted	Investigation workflows; Investigator decision-making; Case management; SAR filing; Compliance decisioning

Fraud Understanding

What is the attack?

Attackers use GenAI to generate convincing communications, evidence, and impersonation attempts targeting AML investigators and compliance officers. By manipulating investigators or the evidence they see, attackers can cause investigations to be closed, risk assessments to be downgraded, and suspicious activity to go unreported. The FATF's October 2025 Plenary highlighted that criminals are increasingly exploiting generative AI, AI agents, and deepfake technologies to undermine trust in financial systems.

Traditional mechanism:

Before GenAI, investigator social engineering required manual communications, generic phishing attempts, and limited personalization. Detection was possible through poor impersonation quality and generic content.

GenAI transformation:

Dimension	Pre-GenAI	GenAI-Transformed
Communication quality	Generic, detectable	Hyper-personalized, professional
Evidence fabrication	Limited, detectable	AI-generated, convincing
Personalization	Manual	AI-powered from OSINT
Scale	Limited	Industrial-scale
Adaptation	Static	Real-time dialogue adaptation

HARD test result: PASS — GenAI creates a qualitatively new capability. The transformation from generic social engineering to hyper-personalized, professional deception is a qualitative leap.

Prerequisites:

- Understanding of AML investigation workflows
- Access to GenAI tools (LLMs, voice cloning)
- Target intelligence (investigator profiles, case details)
- (For voice variant) Voice samples of authority figures

Conceptual attack flow:

1. Attacker identifies AML investigator or compliance officer
2. Attacker gathers intelligence on investigator, case, or process
3. Attacker uses GenAI to generate personalized communication
4. Attacker contacts investigator (email, phone, in-person impersonation)
5. Attacker provides fabricated evidence or instructions
6. Investigator manipulated to close case or downgrade risk
7. Fraud activity continues undetected
8. SAR not filed or filed inaccurately

Consequence:

Suspicious activity goes unreported; fraud continues undetected; inaccurate regulatory filings; enhanced exposure to financial crime.

Detection Intelligence

Signal Category	Observable Signal	Why It Matters	Detection Approach	FP Risk
Documentation	AI-generated text patterns	Indicates deception	NLP; LLM-based detection	High
Documentation	External communication with unusual tone	Indicates impersonation	Anomaly detection	Moderate
Case	Unusual case closure patterns	Indicates manipulation	Case pattern analysis	Low
Case	Downgraded risk assessments	Indicates manipulation	Risk pattern analysis	Low
Temporal	Unusual case processing speed	Indicates urgency manipulation	Temporal analysis	Moderate
Network	Similar manipulation patterns across cases	Indicates coordinated attack	Graph analytics	Low
Investigator	Investigator behavior anomalies	Indicates compromise	Behavioral analytics	Moderate

Simulation Intelligence

Component	Description	Classification
Legitimate entities	Investigator personas with normal decision patterns	Algorithmic
Suspicious entities	Attackers targeting investigators	Agentic
Communications	AI-generated emails; Voice clones; Impersonation scenarios	Agentic
Evidence	Fabricated documents; AI-generated evidence	Agentic + Algorithmic
Case interactions	Investigator-attacker interactions; Evidence submissions	Agentic
Attack labels	Investigator social engineering; Case manipulation	Algorithmic
Mutation parameters	Communication variation; Evidence variation; Timing variation	Algorithmic
Legitimate counterparts	Legitimate compliance communications; Legitimate case closures	Algorithmic
Edge cases	Partial manipulation; Mixed legitimate/manipulated cases	Hybrid

Simulation Type: Agentic

Evidence

Claim	Source	Publisher	Date	Evidence Type	Confidence	Maturity
Criminals increasingly exploit GenAI, AI agents, and deepfakes to enable fraud	FATF October 2025 Plenary	FATF	2025	Government report	VERIFIED	STRONG RESEARCH

Claim	Source	Publisher	Date	Evidence Type	Confidence	Maturity
FATF warns of how criminals can exploit GenAI to undermine trust in financial systems	FATF Horizon Scan	FATF	2025	Government report	VERIFIED	STRONG RESEARCH
Insider threats and social engineering are significant AML risks	ACAMS	ACAMS	2025	Industry analysis	VERIFIED	STRONG RESEARCH

FAMILY AML-006: AI-POWERED AML MODEL POISONING / EVASION

Identity

Field	Value
Attack ID	AML-006
Attack Family	AI-Powered AML Model Poisoning / Evasion
Variants	Training data poisoning; Label flipping; Feature manipulation; Adversarial sample injection; Model evasion through adversarial inputs; Backdoor insertion
Primary Lifecycle Stage	AML / Compliance
Cross-stage Stages	Authorization (score manipulation); Payment Initiation (pattern generation); Account Creation (synthetic data injection)
Attacker	Sophisticated attackers with ML knowledge; Insiders with access to training pipelines; State actors; Organized crime groups with technical capability
Target	AML ML models; Training pipelines; Feature pipelines; Model retraining; Model inference
Objective	Poison AML ML models during training or retraining to degrade detection performance, or generate adversarial inputs that evade trained models
AML Control Targeted	ML-based detection models; Training data integrity; Feature pipelines; Model retraining; Model inference

Fraud Understanding

What is the attack?

Attackers manipulate AML ML models by injecting poisoned data during training or generating adversarial inputs that evade trained models. This degrades model performance over time, reducing detection rates for genuine fraud. Research demonstrates that GAN-based methods can generate samples with properties of malicious activity, moving cumulative amounts through networks of accounts without being detected by existing systems. Just 7% to 8% of poisoned data can degrade the performance of fraud detection, sanctions screening or credit scoring.

Traditional mechanism:

Before GenAI, model poisoning required manual data manipulation or access to training pipelines. The complexity limited the attack to insider threats or highly sophisticated attackers.

GenAI transformation:

Dimension	Pre-GenAI	GenAI-Transformed
Poisoning data generation	Manual, limited	AI-generated poisoned samples at scale
Adaptation	None	Adaptive to model retraining
Scalability	Limited	Industrial-scale
Detection evasion	Detectable as anomalies	Appears as legitimate variation
Automation	Manual	AI-agent automated

HARD test result: PASS — GenAI creates a qualitatively new capability. The transformation from manual, detectable poisoning to adaptive, AI-generated poisoning at scale is a qualitative leap.

Prerequisites:

- Understanding of AML ML model architecture
- Ability to influence training data
- Ability to manipulate transaction inputs (for evasion)
- Access to feature pipelines

Conceptual attack flow (Poisoning):

- 1. Attacker studies AML ML model architecture and training data
- 2. Attacker generates poisoned training data using GenAI
- 3. Poisoned data injected into training pipeline
- 4. Model retrains with poisoned data
- 5. Model performance degrades over time
- 6. Fraud detection rates decline
- 7. Fraud proceeds undetected

Conceptual attack flow (Evasion):

- 1. Attacker understands AML model features and decision boundaries
- 2. Attacker uses GenAI to generate adversarial inputs
- 3. Manipulated transactions submitted
- 4. Model scores transactions as low-risk
- 5. Fraud transactions approved

Consequence:

Systemic detection degradation; increased fraud losses; increased false positives or false negatives; regulatory penalties for detection failures; costly model retraining.

Detection Intelligence

Signal Category	Observable Signal	Why It Matters	Detection Approach	FP Risk
Model	Model performance degradation	Indicates poisoning	Model performance monitoring	Low
Model	Feature distribution shifts	Indicates data manipulation	Distribution monitoring	Moderate
Model	Unnatural label patterns	Indicates label flipping	Label distribution analysis	Low
Data	Anomalous training data patterns	Indicates poisoning	Data quality monitoring	Moderate
Data	Unnatural feature relationships	Indicates adversarial inputs	Feature correlation analysis	Moderate
Temporal	Model performance decline patterns	Indicates poisoning	Temporal model monitoring	Low
Network	Coordinated data injection patterns	Indicates poisoning attack	Graph analytics	Low

Simulation Intelligence

Component	Description	Classification
Legitimate entities	Normal transaction data for training	Algorithmic
Suspicious entities	Poisoned training samples; Adversarial inputs	Algorithmic
Transactions	Data with embedded poisoning; Adversarial patterns	Algorithmic
Account histories	Training data with normal patterns	Algorithmic
Model interactions	Model training; Model inference; Retraining	Agentic
Attack labels	Model poisoning; Adversarial evasion; Data manipulation	Algorithmic
Mutation parameters	Data variation; Feature variation; Timing variation	Algorithmic
Legitimate counterparts	Legitimate data drift; Normal model performance variation	Algorithmic
Edge cases	Partial poisoning; Mixed legitimate/poisoned data	Hybrid

Simulation Type: Algorithmic + Agentic

Evidence

Claim	Source	Publisher	Date	Evidence Type	Confidence	Maturity
7-8% of poisoned data can degrade fraud detection, sanctions screening or credit scoring	Forbes/Research	Forbes	2025	Industry/Research	VERIFIED	STRONG RESEARCH
Data poisoning creates economic losses and compliance exposure	Forbes	Forbes	2025	Industry analysis	VERIFIED	STRONG RESEARCH

Claim	Source	Publisher	Date	Evidence Type	Confidence	Maturity
The GANfather method can move cumulative amounts through networks without detection	Academic research	Research	2025	Academic research	VERIFIED	STRONG RESEARCH
Model poisoning attacks can implant covert decision rules affecting fraud screening	Academic research	Research	2025	Academic research	VERIFIED	STRONG RESEARCH
Adversarial ML attacks can target fraud detection models	Mastercard	Mastercard	2025	Payment network	VERIFIED	STRONG RESEARCH
AI may be used to poison ML training data	FATF Horizon Scan	FATF	2025	Government report	VERIFIED	STRONG RESEARCH

PART 4 – COVERAGE AUDIT

AML/Compliance Surface	Considered?	Covered By	Separate Family?	Decision	Reason
Transaction monitoring evasion	✓	AML-001	No (core)	KEEP	GenAI enables adaptive ML-evasive patterns
Behavioral camouflage	✓	AML-001	No	MERGE → AML-001	Core mechanism of AML-001
Synthetic financial narratives	✓	AML-003	No (core)	KEEP	GenAI enables convincing investigation deception
Synthetic entities	✓	AML-004	No (core)	KEEP	GenAI enables adaptive screening evasion
Documentation manipulation	✓	AML-003	No	MERGE → AML-003	Core mechanism of AML-003
Sanctions screening manipulation	✓	AML-004	No (core)	KEEP	GenAI enables identity variation evasion
Entity resolution manipulation	✓	AML-004	No	MERGE → AML-004	Core mechanism of AML-004
Compliance model attacks	✓	AML-006	No (core)	KEEP	GenAI enables adaptive poisoning/evasion
Adversarial AML behavior	✓	AML-006	No	MERGE → AML-006	Core mechanism of AML-006
Data/input manipulation	✓	AML-006	No	MERGE → AML-006	Core mechanism of AML-006
Alert flooding	✓	AML-002	No (core)	KEEP	GenAI enables industrial-scale alert generation
False-positive generation	✓	AML-002	No	MERGE → AML-002	Core mechanism of AML-002
Investigator manipulation	✓	AML-005	No (core)	KEEP	GenAI enables hyper-personalized deception
Adaptive monitoring evasion	✓	AML-001	No	MERGE → AML-001	Core mechanism of AML-001
Network-level AML evasion	—	—	—	MOVE → Cross-stage	Spans multiple stages/institutions
Agentic interaction with compliance systems	—	—	—	MOVE → AI-Agent Commerce	Primary stage is AI-agent commerce
Mule activity	—	—	—	MOVE → Cash-out/Mule	AML detects; mechanism is mule operation
Money laundering	—	—	—	MOVE → Cash-out/Mule	AML detects; mechanism is laundering
Suspicious transactions	—	—	—	MOVE → Payment Initiation	Transaction fraud belongs elsewhere
Shell company abuse	—	—	—	MOVE → KYC	Primary attack is identity establishment
Beneficial ownership concealment	—	—	—	MOVE → KYC	Primary attack is identity concealment
Insider-facilitated AML control suppression	—	—	—	MOVE → Employee/Insider	Primary attack is insider action
Social engineering of AML investigators	—	—	—	MOVE → Employee/Insider	Primary target is employee
Layering	—	—	—	MOVE → Cross-stage/Network	Spans multiple accounts/stages
Trade-Based Money Laundering	—	—	—	MOVE → Merchant/Cross-stage	Cross-border trade fraud

AML/Compliance Surface	Considered?	Covered By	Separate Family?	Decision	Reason
Legitimate business fronts	—	—	—	MOVE → Merchant	Business creation is merchant-stage
Cryptocurrency laundering	—	—	—	MOVE → Cash-out/Mule	Cash-out mechanism
Hawala	—	—	—	DROP	No GenAI transformation

PART 5 – PRODUCT MAPPING

RED TEAM – What the Attacker/Simulator Generates

Generated Element	Description	Method
Synthetic transaction patterns	Optimized amounts, timing, velocities with natural variation	Algorithmic
Alert-triggering patterns	Patterns designed to trigger specific AML alert types	Algorithmic
Synthetic narratives	AI-generated transaction explanations and justifications	Agentic
Synthetic documents	Invoices, contracts, business agreements, financial statements	Agentic + Algorithmic
Identity variations	Name variations; Entity structures; Beneficial ownership arrangements	Algorithmic + Agentic
Synthetic entities	Complete entity profiles with digital footprints	Agentic
Poisoned training data	Data samples designed to degrade ML models	Algorithmic
Adversarial inputs	Transaction inputs optimized to evade ML models	Algorithmic + Agentic
Investigator communications	Hyper-personalized emails, voice clones, impersonation scenarios	Agentic
Attack labels	Monitoring evasion; Alert flooding; Narrative fabrication; Screening evasion; Investigator deception; Model poisoning	Algorithmic

GENERATE – Classification

Component	Classification	Description
Synthetic transaction patterns	Algorithmic	Parameterized optimization with natural variation
Alert-triggering patterns	Algorithmic	Parameterized patterns for specific alert types
Synthetic narratives	Agentic	LLM-generated explanations with context awareness
Synthetic documents	Agentic + Algorithmic	AI-generated documents with consistent metadata
Identity variations	Algorithmic + Agentic	Parameterized variations; entity structures
Synthetic entities	Agentic	Complete AI-generated entity profiles
Poisoned training data	Algorithmic	Parameterized sample generation
Adversarial inputs	Algorithmic + Agentic	Feedback-driven input optimization
Investigator communications	Agentic	AI-generated personalized deception
Mutation parameters	Algorithmic	Variation across all dimensions

DEFEND – What the Blue Team Detects

Detection Area	What to Detect	Method
Transaction patterns	Amounts near thresholds; Unnatural consistency; Coordinated patterns	Anomaly detection; Sequence modeling; Graph analytics
Alert patterns	Volume spikes; High false-positive rates; Coordinated timing	Volume anomaly detection; Clustering; Temporal correlation
Narratives	AI-generated text patterns; Unnatural consistency; Metadata anomalies	NLP; LLM-based detection; Document forensics; Cross-document consistency

Detection Area	What to Detect	Method
Identity screening	Name variations; Unnatural corporate structures; Concealed connections	Fuzzy matching; Graph analytics; Entity resolution
Investigator behavior	Unusual case closures; Risk downgrades; Behavior anomalies	Case pattern analysis; Behavioral analytics
Model performance	Performance degradation; Feature shifts; Label anomalies	Model performance monitoring; Distribution monitoring; Data quality
Network relationships	Coordinated patterns; Concealed connections; Entity relationships	Graph Neural Network; Clustering

Signal Combination Framework:

- **Transaction-level:** Amounts, timing, velocities, counterparties
- **Entity-level:** Identity attributes, corporate structures, beneficial ownership
- **Behavioral:** Pattern consistency, behavioral drift, response patterns
- **Graph/Network:** Account relationships, entity connections, coordination patterns
- **Document/NLP:** AI-generated text detection, document forensics, consistency scoring

FEEDBACK – Closing the Loop

How a missed detection generates harder synthetic variants:

- Defender fails to detect a generated AML/compliance attack
- System records the evasion pattern (which signals were bypassed)
- Generate module creates variants that exploit the same blind spot:
 - Adjust transaction pattern parameters
 - Modify alert-triggering patterns
 - Change narrative personalization
 - Vary identity generation approaches
 - Adjust poisoning sample characteristics
 - Modify communication styles
- Defender retrains on the new variants
- Harder variants are generated until detection is reliable

How the system discovers AML model blind spots:

- Monitor detection performance across attack families
- Identify families with high false negative rates
- Generate mutations specifically targeting weak detection areas
- Surface to research team for new attack hypotheses

How defender feeds new attack hypotheses back into Generate:

- Defender identifies a new evasion pattern
- Pattern is codified as a new mutation parameter
- Generate module incorporates the parameter into mutation space
- New variants are generated to test the defender
- Cycle repeats

PART 6 – FINAL SUMMARY

Table 1 – FINAL AML / COMPLIANCE TAXONOMY

ID	Attack Family	Variants	GenAI Classification	Simulation Type	Primary Stage	AML Control	Reason for Distinctness
AML-001	AI-Enhanced Adaptive Transaction Monitoring Evasion	Pattern-free movement; Behavioral camouflage; Risk-score manipulation; Adaptive pattern generation; Temporal optimization	PASS	Agentic + Algorithmic	AML/Compliance	Transaction monitoring; Risk scoring; Behavioral analytics	GenAI enables adaptive, ML-evasive transaction patterns—qualitative leap from manual rule evasion

ID	Attack Family	Variants	GenAI Classification	Simulation Type	Primary Stage	AML Control	Reason for Distinctness
AML-002	AI-Powered AML Alert Flooding / DoS	False-positive flooding; Alert system overload; Investigator overwhelm; Synthetic alert generation	PASS	Agentic + Algorithmic	AML/Compliance	Alert generation; Triage; Case management	GenAI enables industrial-scale, adaptive alert generation that consumes investigative capacity—qualitative leap from manual flooding
AML-003	AI-Generated Synthetic AML Narrative / Evidence	Synthetic narratives; Fabricated documentation; AI-generated justifications; Source of funds fabrication	PASS	Agentic	AML/Compliance	Investigation workflows; CDD/EDD; Source of funds	GenAI enables internally consistent, context-aware narratives that pass investigation—qualitative leap from generic explanations
AML-004	GenAI-Enhanced Sanctions / PEPs Screening Evasion	Identity obfuscation; Name variation generation; Entity concealment; Beneficial ownership concealment	PASS	Algorithmic + Agentic	AML/Compliance	Sanctions screening; PEP screening; Entity resolution	GenAI enables adaptive identity variation and entity structures that evade screening—qualitative leap from manual evasion
AML-005	AI-Powered AML Investigator Social Engineering	Investigator impersonation; AI-generated investigation responses; Case manipulation; Evidence fabrication	PASS	Agentic	AML/Compliance	Investigation workflows; Investigator decision-making	GenAI enables hyper-personalized deception of investigators—qualitative leap from generic social engineering
AML-006	AI-Powered AML Model Poisoning / Evasion	Training data poisoning; Label flipping; Adversarial sample injection; Model evasion; Backdoor insertion	PASS	Algorithmic + Agentic	AML/Compliance	ML-based detection; Training pipelines; Feature pipelines	GenAI enables adaptive, scalable poisoning that degrades detection systems—qualitative leap from manual attacks

Table 2 – ATTACK × SIGNAL

Attack Family	Transaction	Account	Entity	Network	Behavioral	Temporal	Document	Screening	Model
AML-001: Monitoring Evasion	Optimized patterns; Unnatural consistency	Consistent with profile	Business profiles too perfect	Coordinated patterns	Too-perfect consistency	Unnatural timing consistency	N/A	N/A	Risk score stability
AML-002: Alert Flooding	Patterns triggering alerts	Multiple accounts	N/A	Coordinated attack patterns	Unnatural variety	Coordinated timing	N/A	N/A	Alert volume anomaly
AML-003: Narrative Fabrication	N/A	Account history	N/A	Similar narrative patterns	Response patterns	Response timing	AI-generated text; Synthetic documents	N/A	N/A
AML-004: Sanctions Evasion	N/A	Synthetic entities	Unnatural structures	Concealed connections	N/A	Entity creation timing	Fabricated documentation	Name variations	Watchlist match anomalies
AML-005: Investigator Manipulation	N/A	Multiple cases	Investigator behavior	Similar manipulation patterns	Investigator anomalies	Case processing timing	AI-generated communications	N/A	Case outcome anomalies
AML-006: Model Poisoning	Feature anomalies	N/A	N/A	Coordinated data injection	N/A	Performance decline timing	N/A	N/A	Performance degradation

Table 3 – ATTACK × SIMULATION

Attack Family	Entities	Accounts	Transactions	Graph	Timeline	Documents	Alerts	Behavior	Simulation Type
AML-001	Account personas	Optimized patterns	Optimized amounts/timing	Coordination patterns	Pattern evolution	N/A	Low alert rate	Pattern consistency	Agentic + Algorithmic
AML-002	Synthetic entities	Multiple accounts	Alert-triggering	Coordinated patterns	Alert generation	N/A	Volume spikes	Pattern variety	Agentic + Algorithmic
AML-003	Account personas	Under investigation	Transaction patterns	N/A	Investigation timeline	Synthetic narratives/documents	N/A	Response patterns	Agentic
AML-004	Synthetic entities	Concealed entities	N/A	Concealed connections	Entity creation	Fabricated documents	Low hit rate	N/A	Algorithmic + Agentic
AML-005	Investigator personas	Case histories	N/A	Manipulation patterns	Investigation timeline	Communications	Case anomalies	Investigator behavior	Agentic
AML-006	N/A	Training data	Poisoned/adversarial	Data injection	Model degradation	N/A	Detection decline	Model performance	Algorithmic + Agentic

Table 4 – TAXONOMY DECISIONS

Candidate	Decision	Reason
AML-001: Adaptive Transaction Monitoring Evasion	KEEP	Distinct mechanism; PASS test; AML-primary
AML-002: Alert Flooding / DoS	KEEP	Distinct mechanism; PASS test; AML-primary
AML-003: Synthetic Narrative / Evidence	KEEP	Distinct mechanism; PASS test; AML-primary
AML-004: Sanctions/PEPs Screening Evasion	KEEP	Distinct mechanism; PASS test; AML-primary
AML-005: Investigator Social Engineering	KEEP	Distinct mechanism; PASS test; AML-primary
AML-006: Model Poisoning / Evasion	KEEP	Distinct mechanism; PASS test; AML-primary
Structuring/Smurfing	MERGE → AML-001	Variant of adaptive pattern evasion
Low-and-Slow	MERGE → AML-001	Variant of adaptive pattern evasion
Behavioral Camouflage	MERGE → AML-001	Core mechanism of AML-001
False-Positive Generation	MERGE → AML-002	Core mechanism of AML-002
Alert System Probing	MERGE → AML-002	Variant of alert system manipulation
Fabricated Source of Funds	MERGE → AML-003	Variant of narrative/evidence fabrication
Name Variation Exploitation	MERGE → AML-004	Core mechanism of AML-004
Data Poisoning	MERGE → AML-006	Core mechanism of AML-006
Adversarial Evasion	MERGE → AML-006	Core mechanism of AML-006
Insider-Facilitated AML Control Suppression	MOVE → Employee/Insider	Insider action; not AML-primary
Social Engineering of AML Investigators	MOVE → Employee/Insider	Employee target; not AML-primary
Layering	MOVE → Cross-stage/Network	Spans multiple stages
Cross-Institution Fragmentation	MOVE → Cross-stage/Network	Spans multiple institutions
Cuckoo Smurfing	MOVE → Cash-out/Mule	Mule recruitment mechanism
Cryptocurrency Laundering	MOVE → Cash-out/Mule	Cash-out mechanism
Shell Company Abuse	MOVE → KYC	Primary attack is identity/entity establishment
TCSP Abuse	MOVE → KYC	Primary attack is identity/entity establishment
Beneficial Ownership Concealment	MOVE → KYC	Primary attack is identity concealment
Trade-Based Money Laundering	MOVE → Merchant/Cross-stage	Cross-border trade fraud
Legitimate Business Fronts	MOVE → Merchant	Business creation is merchant-stage
Suspicious Transactions	MOVE → Payment Initiation	Transaction fraud belongs elsewhere
Hawala	DROP	No GenAI transformation

FINAL METRICS

Metric	Value
Total final attack families	6
Total variants	18+
Important candidates merged	Structuring → AML-001; Low-and-Slow → AML-001; Behavioral Camouflage → AML-001; False-Positive Generation → AML-002; Alert System Probing → AML-002; Fabricated Source of Funds → AML-003; Name Variation Exploitation → AML-004; Data Poisoning → AML-006; Adversarial Evasion → AML-006
Important candidates moved	Insider-Facilitated AML Control Suppression → Employee/Insider; Investigator Social Engineering → Employee/Insider; Layering → Cross-stage/Network; Cross-Institution Fragmentation → Cross-stage/Network; Cuckoo Smurfing → Cash-out/Mule; Cryptocurrency Laundering → Cash-out/Mule; Shell Company Abuse → KYC; TCSP Abuse → KYC; Beneficial Ownership Concealment → KYC; Trade-Based Money Laundering → Merchant/Cross-stage; Legitimate Business Fronts → Merchant
Important candidates dropped	Hawala (no GenAI transformation)
Theoretical/future threats	AI-generated regulatory filing manipulation; Autonomous AI agent AML evasion; Cross-institution coordinated evasion; Real-time adversarial model feedback; Agentic compliance workflow manipulation
Evidence gaps	Real-world frequency of AI-generated AML narratives in production; Specific detection evasion effectiveness against production AML systems; Impact of emerging GenAI architectures on AML model poisoning
AML/Compliance surfaces covered	Transaction monitoring; Suspicious activity detection; Behavioral monitoring; Alert generation; Alert triage; Investigation workflows; Case management; Sanctions screening; PEP screening; Entity resolution; Risk scoring; CDD/EDD; Source of funds verification; Regulatory reporting; ML-based detection; Training pipelines; Feature pipelines; Watchlist matching; Beneficial ownership verification
Important AML surfaces not represented separately	Mule activity (moved to Cash-out/Mule); Money laundering (moved to Cash-out/Mule); Suspicious transactions (moved to other stages); Trade-based money laundering (moved to Merchant/Cross-stage); Business fronts (moved to Merchant)
Lifecycle stages that absorb moved attacks	Employee/Insider; Cross-stage/Network; Cash-out/Mule; KYC; Merchant; Payment Initiation

UNIFYING THEME

The unifying theme across all 6 AML/Compliance families is the shift from **detectable anomalies** to **indistinguishable operational reality**.

Fraudsters no longer simply avoid detection—they use GenAI to:

- **Generate patterns that appear legitimate** (AML-001): Transaction patterns optimized to evade ML-based monitoring
- **Weaponize the detection system itself** (AML-002): Flooding AML systems with synthetic alerts that consume investigative capacity
- **Create internally consistent narratives** (AML-003): AI-generated explanations that pass investigation scrutiny
- **Evolve identities that evade screening** (AML-004): Adaptive variations that bypass sanctions and PEP screening
- **Manipulate the human layer** (AML-005): Hyper-personalized deception of AML investigators
- **Degrade the detection models** (AML-006): Poisoning ML models through adversarial data injection

The core vulnerability:

AML/Compliance systems are designed to detect **obvious** suspicious activity. GenAI makes fraud:

- **Pattern-free**: No detectable structure to monitor
- **Context-aware**: Consistent with legitimate activity
- **Adaptive**: Evolves to avoid detection
- **Scalable**: Industrial-scale evasion
- **Indistinguishable**: Appears identical to legitimate activity

The attack surface has shifted from:

"Can we avoid detection?" → Traditional AML evasion

To:

"Can we generate an entirely parallel financial reality that detection systems accept as legitimate?"

Fraudsters no longer rely on:

- Manual structuring
- Simple threshold avoidance
- Generic explanations
- Detectable forgeries

They use GenAI to generate:

- Optimized transaction patterns that appear legitimate
- Synthetic alerts that overwhelm investigation capacity
- Internally consistent narratives that satisfy investigators
- Adaptive identities that evade screening
- Poisoned data that degrades detection systems

The fundamental shift:

The AML system itself is no longer just a **detection mechanism**—it is an **attack surface**. The attacker's objective is not merely to avoid detection but to manipulate the detection system itself, whether by:

- **Evading** its monitoring (AML-001)
- **Overwhelming** its capacity (AML-002)
- **Deceiving** its investigators (AML-003, AML-005)
- **Bypassing** its controls (AML-004)
- **Degrading** its models (AML-006)

This represents a fundamental change in the threat model: AML systems are not just defenses against fraud; they are **targets** of GenAI-powered attacks.

END OF FINAL AML / COMPLIANCE TAXONOMY AUDIT + MASTER FRAUD INTELLIGENCE DOCUMENT